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Dataset description and background: Phishing websites are a major cybersecurity threat, and detecting them from URL characteristics alone allows fast, low-cost screening before a page is ever loaded. The dataset contains labeled phishing and benign URLs. Size: [X rows, Y columns from the Kaggle page]. If we use the Data in Brief dataset, it has 111,660 rows and 26 columns.

Problem type: Binary classification (phishing vs. legitimate).

Dataset link: [Kaggle link]. Backup: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12552964/

Target: Categorical, 2 classes: 0 = benign, 1 = phishing.

Predictor variables: Mostly quantitative lexical and structural URL features (URL length, domain length, digit ratio, entropy, HTTPS usage, and similar), plus categorical or text fields (domain, TLD, raw URL) that we can encode or engineer features from.

Planned approach: Data cleaning, scaling, and pipelines. Class-imbalance handling if needed, since the backup dataset is about 90% benign. Compare logistic regression, SVM (different kernels), decision tree, and random forest, with mini-batch GD where applicable. Grid search for tuning.

Evaluation metrics: Precision, recall, F1, and ROC AUC, not just accuracy. With imbalanced classes, accuracy alone can hide poor phishing detection, and a missed phishing site is costlier than a false alarm.

Desired performance (needed for your conclusion file): For the Data in Brief dataset, the authors report accuracy above 96% and ROC AUC above 0.99 for both a Random Forest baseline and a MiniLM embedding with Logistic Regression. That gives you a citable benchmark to aim for.